How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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Abstract

Spatial memory in primates is not a sensor-independent module. Hippocampal cells code joint place–view–action state in part because the primate eye samples the world through a foveated sensor that resolves a small high-acuity region per glance; rodents with near-panoramic vision instead develop classic place fields, and individuals deprived of vision recruit hippocampal circuits to compensate. The biological pattern is consistent: *the structure of the sensor shapes the structure of spatial memory*. Whether the same holds for artificial navigation agents — whose recurrent memories also develop cognitive-map-like representations under task pressure alone — has not been tested directly, because prior work has either removed vision entirely or used spatially uniform input, never varying sensor structure as an independent variable. We isolate the principle in a controlled silicon model: five navigation agents are trained with everything but the visual pathway held identical, spanning a sensor-structure spectrum from no vision through a resolution-collapsed encoder, foveated vision (with fixed and learned gaze), and full uniform vision. Probing their recurrent state reveals an *encoder–memory race*: the recurrent state is forced to maintain a world-frame spatial code when the visual encoder cannot supply enough spatial-feature variety per step, and lets the code be overwritten when the encoder is rich enough to substitute. Independently, agents that solve the same task at comparable competence develop hidden states in linearly disjoint subspaces, and cross-condition memory transplants pay a task cost that grows with sensor-format mismatch. The often-asserted intuition that "richer perception begets richer memory" is wrong in this regime: the relationship is inverse, and encoder capacity becomes a candidate training-time lever for inducing cognitive-map-like internal representations. The agent pattern echoes — without depending on — biological findings on hippocampal compensation under sensory deprivation and cross-modality remapping, suggesting a sensor–memory coupling principle whose generality across architectures is open.

1 Introduction

Primate spatial memory is shaped by how primates see. Hippocampal cells code joint place–view–action state in part because the primate eye samples the scene through a foveated sensor [Wirth et al., 2017, Rolls, 2024]; rodents with near-panoramic vision instead show classic place fields. Sensor structure also reorganises memory *within* a single animal — bats produce non-overlapping hippocampal populations under vision vs. echolocation [Geva-Sagiv et al., 2016], and congenitally blind humans develop enhanced hippocampal coupling on non-visual spatial tasks [Kupers and Ptito, 2014]. The biological pattern is consistent: *the structure of the sensor reshapes the structure of spatial memory*. We ask whether the same is true of artificial navigation agents.

The emergent-cognitive-maps literature has shown that map-like structure arises in the recurrent state of navigation agents trained with deep RL, including agents with no vision at all [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018]. These results establish that *some* map-like structure is learned without being designed in. They leave open the complementary question we take up here: holding task and architecture fixed, does the structure of the visual sensor reshape *what* recurrent memory encodes? We isolate sensor structure as the only varied component across five conditions spanning a sensor-structure spectrum (Figure 1): no vision, a resolution-collapsed encoder, foveated vision with fixed gaze, foveated vision with learned gaze, and uniform full-resolution vision. The resolution-collapsed control separates "no vision" from "vision the encoder cannot resolve"; foveation occupies a biologically motivated point in between.

We test three hypotheses. *H1 (encoder–memory race)*: when the visual encoder supplies little spatial-feature variety per step, the recurrent state should compensate by carrying a world-frame spatial code; richer encoder output should let memory rely on visual features instead. *H2 (format divergence)*: different sensor conditions should produce hidden-state representations in distinct linear subspaces, not scaled versions of a common code. *H3 (gaze location)*: within the rich-encoder regime, where a foveated sensor is centred should act as a second content axis independent of the sensor transform. Each has a direct biological analog (§5.2).

Three findings emerge with the strongest evidence. An encoder–memory race along the sensor-structure spectrum (H1) that lives inside the recurrent state, not at the encoder. The five agents solve the task at comparable competence but develop hidden states in linearly disjoint subspaces, with measurable cross-condition memory-transplant cost (H2). And probe-readable vs. policy-used memory dissociates in opposite directions for two specific conditions, suggesting that "has a cognitive map" and "uses a cognitive map" need not coincide. Foveation under our blur model converges with uniform on H1 but diverges on H2, behaviour, and dataset shift. H3 (gaze location) is tested by a foveated-shifted control in training at submission.

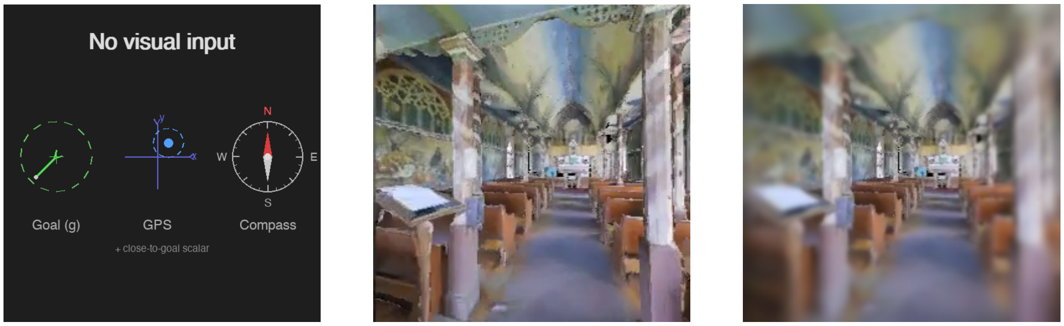
The cross-condition pattern is correlation-level; a scaling-sweep experiment that would test the H1 mechanism causally is in progress (Appendix H). Foveation-specific predictions involving alternative foveation models depend on experiments still in training (§4.4, Appendix G). Each finding has a direct biological precedent which we discuss as convergent (not conclusive) evidence in §5.2.

2 Related Work

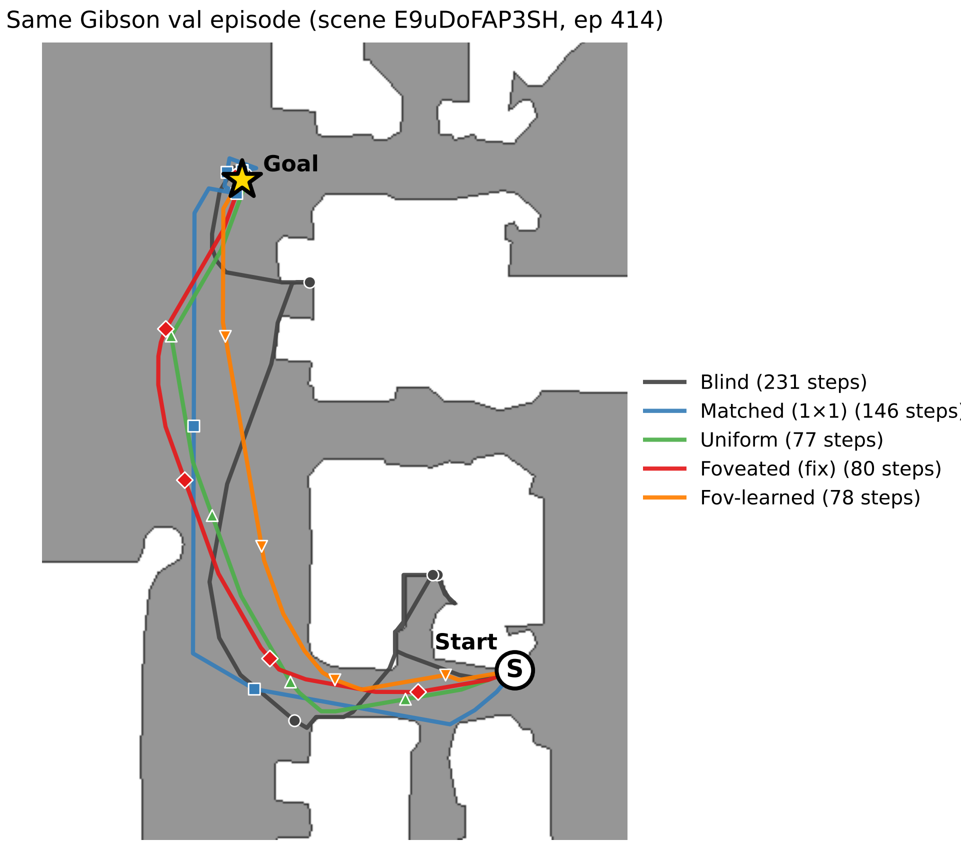
Cognitive maps in RL navigation agents Wijmans et al. [Wijmans et al., 2023] demonstrated that map-like representations emerge in the recurrent memories of DD-PPO agents even without visual input; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like units in recurrent networks trained on path integration; Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use either absent or spatially uniform perception; none vary input quality spatially, and none test whether different sensors produce different representational formats.

Foveated vision and gaze policies in deep learning Foveated rendering [Strasburger et al., 2011, Rosenholtz, 2016] exploits eccentricity-dependent acuity falloff for efficiency. Learned glimpse policies appear in recurrent attention models [Mnih et al., 2014]; Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies following the Sprague & Ballard [Sprague and Ballard, 2003] framework. These address visual search or task performance; none asks how sensor structure reshapes the learned spatial memory of a downstream navigation policy.

Probing neural representations Belinkov [Belinkov, 2022] surveys probing classifiers; Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorisation; Kornblith et al. [Kornblith et al., 2019] introduced linear CKA for cross-model similarity. We adopt linear probing, the Hewitt–Liang selectivity metric, and unaligned linear CKA, and complement them with two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery) to validate the probe-based findings outside the linear-probe framework.



(a) Blind (b) Uniform (c) Foveated (fixed gaze)



(d) The 5 conditions on the same MP3D-train episode (scene E9uDoFAP3SH, ep 40422 in the env iterator; same paired index 414 across our probing data), trajectories overlaid on the Habitat-rendered top-down occupancy map. Each condition's SPL on this episode is within ± 0.10 of its per-condition median, so all five are typical successful runs. Bottleneck conditions (blind 231 steps, matched 146) take longer paths through the corridor; rich-encoder conditions (uniform / foveated / fov-learned, 77–80 steps) take near-direct paths along the same corridor.

Figure 1: The five visual conditions differ only in what reaches the agent's encoder, but produce visibly different navigation behaviours. *Inputs:* (a) Blind: no visual input; only the non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar) — the Wijmans [Wijmans et al., 2023] configuration. (b) Uniform: 256×256 full-resolution RGB. (c) Foveated (fixed gaze): eccentricity-dependent Gaussian blur centred on the image middle ($\sigma_{\max} = 8$, quadratic falloff). Two additional conditions not pictured: *matched-compute* (a down-sampled uniform view at 48×48, whose total pixel budget matches foveated and whose ResNet-18 encoder produces a 1×1 feature map) and *foveated (learned gaze)*, which adds a learned (x, y) gaze offset to (c). *Behaviour:* (d) all five conditions solve the same PointGoal navigation task; the trajectory shape and length differ systematically, previewing the encoder–memory race finding (§4.2).

What is new here Three things distinguish this work. (i) A *controlled* five-condition ablation with everything but the visual pathway held identical, including the matched-compute condition that decouples "no vision" from "vision the encoder cannot resolve". (ii) A protocol-corrected probing methodology (deterministic action selection at probe time; stochastic sampling inflates probe R^2 via a 4-step quasi-static-trajectory artefact, Appendix A). (iii) Convergent behavioural validation via memory transplant and shortcut discovery, outside the linear-probe framework. Biological motivation and the parallel to animal-navigation research are integrated in §5.2 rather than recapitulated here.

3 Methods

3.1 Setup

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes; evaluation uses 18 held-out MP3D test scenes (1008 episodes). All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar), each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions train to ∼250M environment frames; the blind baseline trains to 342M to allow slower convergence from proprioception alone.

The five visual conditions (Figure 1): *blind* (no visual input); *uniform* (full 256×256 RGB encoded by ResNet-18); *foveated (fixed)* (eccentricity-dependent Gaussian blur, $\sigma_{\max} = 8$ quadratic falloff, gaze locked at image centre, same ResNet-18 encoder); *matched-compute* (48×48 uniform RGB; at this resolution ResNet-18's 32-fold spatial downsampling collapses the feature map to 1×1, removing spatial structure but preserving channel information); *foveated (learned)* (same blur model, gaze centre predicted by a lightweight MLP trained end-to-end). Matched-compute is the spatial-bandwidth lower bound for our encoder; it has visual input but its encoder output cannot resolve world-frame position. The 48×48 input also has lower spatial resolution than uniform's 256×256, so this single condition cannot fully separate the 1×1 collapse from low input resolution; the encoder-resolution scaling sweep (Appendix H) addresses this directly. The foveated conditions disable the running-mean-and-variance input normaliser to avoid an in-place autograd conflict along the gaze-decoder gradient path; an invariance control is in training. We also discovered and patched a silent NaN-gradient corruption mode in DD-PPO (Appendix F).

3.2 Probes

After training, all weights are frozen. We collect per-step top-layer LSTM hidden states (512-d) paired with ground-truth labels from $n{=}500$ Gibson held-out episodes under *deterministic* argmax-action rollouts (matching our shortcut / transplant / standard eval protocols; the alternative stochastic protocol gives trivial $\sim$ 4-step rollouts and is documented in Appendix A). We train Ridge probes ($\alpha = 10$) with episode-level 5-fold cross-validation, report mean $\pm\sigma$ across folds, and verify probe specificity with Hewitt–Liang label-permutation selectivity [Hewitt and Liang, 2019]. Probed variables: GPS position, compass heading, distance-to-goal, lag-k path history ($k \in \{0, \dots, 20\}$), visited-region occupancy, full occupancy grid, per-unit spatial information, ego-frame goal vector. For H2 representational geometry we additionally use linear CKA [Kornblith et al., 2019] and cross-condition probe transfer.

We complement probes with two interventions on the LSTM state. *Memory transplant*: paste a donor's hidden state into the recipient at step 30 and compare baseline / self-transplant / cross-transplant SPL across 150 episodes per pair. *Shortcut discovery*: run paired same-scene-different-goal episodes (20 scenes $\times$ 10 pairs) and compare second-episode SPL under reset vs. persistent LSTM. For foveated-fixed, all analyses use ckpt.36 at ∼174M frames (last clean intermediate before the silent-corruption window).

4 Results

4.1 Five conditions, same task

Table 1 collects the headline per-condition numbers. All sighted conditions solve PointGoal at 96–99% success; blind reaches 93% at a longer training budget, reproducing Wijmans et al. [Wijmans et al., 2023]. Success is tightly bunched, so a skill-gap account for the encoded-content differences below is unlikely. The probe is calibrated: distance-to-goal decodes everywhere ($R^2 \in [0.81, 0.90]$); Hewitt–Liang label-permutation selectivity is high precisely where GPS is high (Appendix A).

Table 1: Per-condition headline. Probe R^2 columns are the 5-fold episode-level CV mean $\pm$ std on full-length deterministic-rollout trajectories (no per-episode step cap). *GPS* and *Compass* are world-frame. The bottleneck-vs-rich-encoder gap is in temporal *stability* of the code, not its mid-episode existence (Figure 2b): all five conditions decode top-layer GPS at $R^2 \in [0.6, 0.95]$ in the typical-episode window (steps 50–200); the rich-encoder long-tail collapse is what produces the sub-chance numbers below. Bold marks the H1-relevant ordering (blind $>$ matched $\gg$ rich-encoder).

Condition	Frames	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	342M	0.59	0.93	**+0.95±0.02**	**+0.81±0.08**
Matched-compute	250M	0.67	0.98	**+0.78±0.10**	**+0.64±0.10**
Uniform	250M	**0.85**	**0.99**	−0.31±0.86	+0.36±0.23
Foveated (fix)	174M	0.78	0.96	+0.06±0.88	+0.07±0.69
Fov-learned	250M	0.82	0.98	−2.43±3.98	−1.34±3.14

4.2 Encoder–memory race: a temporal-stability claim (H1)

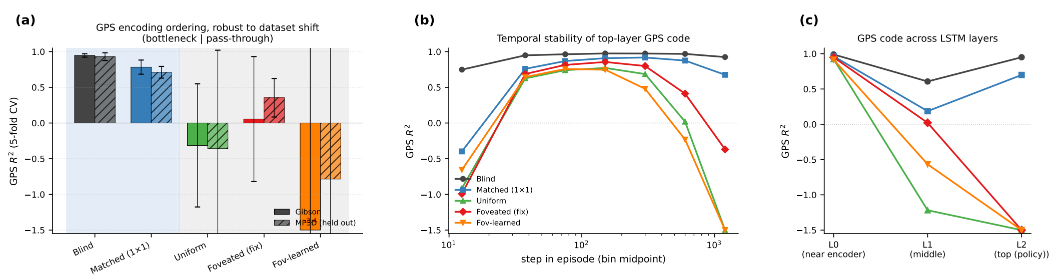


Figure 2: H1 evidence in three panels. *(a)* GPS R^2 (5-fold CV, Ridge $\alpha = 10$, full-length deterministic rollouts) on Gibson and held-out MP3D (300 episodes/condition; same checkpoints, no re-training). Blind and matched-compute (bottleneck conditions, blue background) decode world-frame GPS robustly on both datasets; uniform / foveated / foveated-learned (rich-encoder, grey background) do not, except for foveated recovering to a mild positive ($R^2 = +0.35$) on MP3D. The Compass and DtG numbers are in Table 1 (compass shows the same ordering at smaller magnitude; DtG decodes everywhere because every agent must track it; Appendix D for the companion bar plots). *(b)* Top-layer GPS R^2 binned by step-in-episode (out-of-fold prediction): all five conditions encode top-layer GPS in the typical-episode window (steps $\sim$ 25–200); the rich-encoder conditions then *overwrite* the code (negative R^2 in the long-tail $800+$ bin), while bottleneck conditions hold $R^2 > 0.7$ to step $800+$. The headline "rich-encoder $\approx$ chance" in panel (a) is therefore primarily long-tail behaviour. *(c)* Per-LSTM-layer GPS R^2 (single-fit on top-layer hidden states): all five conditions have GPS at Layer 0 (where the GPS sensor embedding enters the LSTM input vector — this lower-bound is at least partly a sensor-pass-through effect, not the network "computing" position); the divergence is at Layer 2, the policy readout.

Visual-encoder capacity inversely tracks LSTM spatial encoding across our five conditions: bottleneck conditions (blind, matched) encode GPS / compass at the top layer; rich-encoder conditions (uniform, foveated, foveated-learned) decode at chance (Figure 2a, Table 1). DtG decodes robustly in every condition because every agent must track it. **[TODO: Whether the correlation reflects causation is what the §4.4 experiments and the Appendix H resolution sweep test; multi-seed gap-fills are in progress.]**

Temporal stability is the precise claim The no-cap probe conflates "does the LSTM ever encode GPS" with "does it maintain the encoding across episode duration." Step-binned out-of-fold probing separates them (Figure 2b): all five conditions encode top-layer GPS in the typical-episode window; rich-encoder conditions then *overwrite* the code (chance by step 400–800, deeply negative in the long-tail bin), while bottleneck conditions hold it stable to step 800+. The Table 1 "rich-encoder ≈ chance" headline is therefore long-tail behaviour, not mid-episode. *H1 is most precisely a temporal-stability claim about the linearly-decodable top-layer GPS code.* A lag-k probe corroborates: blind GPS decodes at $R^2 = 0.92$ at $k = 20$, matched > 0.5; rich-encoder conditions are at chance for every $k \leq 10$.

Layer specificity and non-linear probe. All five conditions have GPS at LSTM Layer 0 because the GPS-sensor embedding enters there (Figure 2c). The H1 distinction is what happens through Layers 1–2: bottleneck retain to Layer 2, rich-encoder discard. Since the DD-PPO action head is a linear projection from $\mathbf{h}_2$, this maps to "policy-accessible". A 2-layer MLP on the same top-layer states recovers GPS for rich-encoder conditions ($R^2 \in [0.51, 0.73]$); H1 is therefore precisely about *linear* top-layer decodability, not whether *any* probe can recover position. The MLP result alone cannot exclude exploitation of position-correlated features (wall distance, scene identifiers); Appendix B discusses both. The bottleneck-vs-pass-through partition is preserved on held-out MP3D (Figure 2a); two single-seed cross-dataset shifts (foveated GPS chance $\rightarrow +0.35$; foveated-learned compass $-1.34 \rightarrow +0.41$) await multi-seed replication.

Proposed mechanism We *propose* the following account, consistent with the data but not directly tested by a causal intervention. In PointNav, GPS and compass enter the LSTM input at Layer 0 directly, via a 32-d sensor embedding concatenated with the visual encoder's flattened output (Figure 2c shows GPS at Layer 0 across all five conditions, $R^2 > 0.91$). The encoder feature-map probe (§4.4, Figure 4) shows that none of the three sighted encoders linearly re-derive GPS from their visual stream: matched $R^2 = -3.14$, uniform -0.32, foveated -0.65. The encoder–memory race is therefore not about whether the encoder *re-derives* position; it is about how much *visual feature variety* the encoder hands to the LSTM, and whether that variety is enough to substitute for integrating the Layer-0 GPS-sensor signal across time. Bottleneck conditions (blind, no encoder; matched-compute, 1×1 spatial output $\rightarrow$ minimal feature variety per step) leave the LSTM with little visual structure to substitute, so it integrates the Layer-0 GPS signal across time and *maintains* a top-layer linearly-readable GPS code. Rich-encoder conditions (uniform / foveated / foveated-learned, 8×8 spatial output $\rightarrow$ many position-correlated visual features) give the LSTM enough visual variety to base actions on those features instead, and the top-layer GPS code drifts off as the episode progresses. Task success is tightly bunched, so a skill-gap artefact is unlikely to explain the encoding gap. The temporal dynamic in Figure 2b (rich-encoder GPS code emerging early at Layer 0 then decaying through Layers 1–2 to chance at the top layer) is the direct fingerprint of this account. **[TODO: A direct causal test of the mechanism — ablating the visual stream mid-rollout in rich-encoder agents to see whether the top-layer GPS code re-emerges, or perturbing the GPS sensor input mid-rollout in bottleneck agents to see whether SPL collapses — is left for future work.]**

Relation to Wijmans et al. [Wijmans et al., 2023]. They showed blind agents develop spatial representations; our blind result ($R^2 = 0.95$) replicates this and matched-compute extends it to a condition that has visual input but 1×1 spatial encoder output. The shared trigger is therefore minimal visual-feature variety per step, not absence of vision. The parallel to animal-navigation findings is treated separately as convergent (not mechanistic) evidence in §5.2.

4.3 Format-level divergence across conditions (H2)

Behavioural test (memory transplant) The most direct test of "do these representations matter?" is to paste a donor's mid-episode hidden state into a recipient's policy at step 30 and let the recipient drive the rest (Figure 3, right). The isolated condition-mismatch cost — cross-transplant SPL minus recipient's self-transplant SPL — shows three patterns. *Asymmetry.* Blind$\rightarrow$uniform incurs -0.38 SPL while uniform$\rightarrow$blind is neutral within self-noise; bottleneck-donor states hurt rich-encoder recipients far more than the reverse. *Recipient ranking.* Uniform suffers most from *any* foreign donor (column mean ~ -0.20); blind least (~ -0.02 to -0.05). Blind tolerates foreign donors; rich-encoder recipients do not. *Order-of-magnitude probe-similarity blind spot.* Pairs that are indistinguishable under linear similarity (CKA $< 10^{-4}$ across the board) differ by 0.01–0.38 SPL

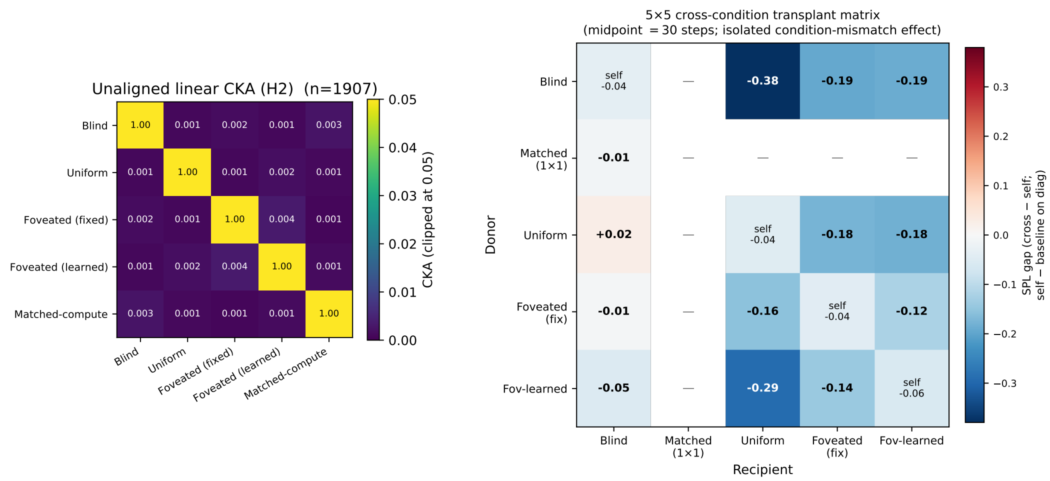


Figure 3: H2 convergent evidence, two views. *Left:* Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states (n=95,640 steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4}. *Right:* 5×5 cross-condition memory-transplant matrix at midpoint t=30. Cells off-diagonal report cross-transplant SPL minus the recipient's self-transplant SPL (the isolated condition-mismatch component, with rollout-divergence mechanics subtracted); diagonal cells report self-transplant SPL minus baseline (the rollout-divergence noise floor, ≈ -0.04 to -0.06). The matrix is asymmetric: blind→uniform incurs -0.38 SPL while uniform→blind is essentially neutral ($+0.02$); foveated-learned→uniform -0.29, blind→foveated -0.19. *Pairs that are indistinguishable to linear similarity (CKA noise floor) differ markedly under transplant.* Cells with '—' are not yet measured (matched as recipient, jobs in flight). A midpoint-sweep view of three representative pairs across midpoints $\{0, 15, 30, 60, 90\}$ confirming the noise-floor and stability arguments is in Appendix D (Figure 8).

Table 2: Probe-transfer GPS R^2 on deterministic-rollout hidden states: train on row condition, test on column. Self-accuracy diagonal is 0.89–0.99; every cross-condition transfer is catastrophically negative ($R^2 \ll -800$, often $\ll -10^4$), reflecting that a probe trained on one condition's hidden-state geometry does not just degrade on another — it predicts off-manifold values.

Train \ Test	Blind	Uniform	Fov-fix	Fov-lrn	Matched-compute
Blind	$+$**0.99**	-844	$-11{,}301$	$-10{,}731$	$-55{,}706$
Uniform	$-1{,}383$	$+$**0.90**	$-7{,}046$	$-3{,}305$	$-110{,}317$
Fov-fix	$-2{,}091$	$-2{,}792$	$+$**0.92**	$-1{,}263$	$-39{,}100$
Fov-lrn	$-6{,}722$	$-2{,}423$	$-6{,}205$	$+$**0.89**	$-38{,}842$
Matched-compute	$-10{,}621$	$-4{,}832$	$-4{,}984$	$-58{,}169$	$+$**0.96**

under transplant. **[TODO: 4 matched-recipient cells of the 5×5 are queued; asymmetry will be tested for completeness on landing.]**

Convergent linear-similarity evidence. Three further measures point to the same "different format" conclusion. *Pairwise CKA* [Kornblith et al., 2019] at the noise floor across all condition pairs (Figure 3, left; within-condition split-half CKA ≈ 1, so the near-zero off-diagonals are not estimator noise). *Cross-condition probe transfer* (Table 2) is catastrophically negative in every off-diagonal cell, so a probe trained on one condition's hidden states predicts off-manifold values on another. *1-NN purity*: among $1500\times5 = 7500$ pooled top-layer states, every sample's Euclidean nearest neighbour shares its condition (purity 1.000 vs. chance 0.20); a t-SNE projection (Appendix D) confirms this visually. **[TODO: 1-NN at larger sample size and under domain shift (MP3D / fov-shifted) is a further-investigate item.]**

Ruling out alternatives. Probe capacity is not the bottleneck (self-probes succeed everywhere); sample-size asymmetry is ruled out by common-sample truncation; task competence is bunched (96–99%), so the divergence is not a skill difference; probe transfer fails symmetrically, ruling out one condition's code simply embedding another's.

Boundaries. Our similarity tests are linear or near-linear; non-linear measures could reveal higher-order shared structure. Our five sensors are diverse but not exhaustive; hybrid sensors (sharp fovea + coarse periphery, depth-only) might produce intermediate geometries.

4.4 Foveation: convergence on H1, divergence on H2 + behaviour

Foveated-fixed and uniform fall in the same rich-encoder pass-through regime on the headline H1 metric (top-layer GPS R^2 at chance), but they are not interchangeable on H2 / behavioural / dataset-shift axes (Table 3). The take-away: *within the rich-encoder regime defined by H1, foveation and uniform produce internally distinct representations whose differences only the H2-and-beyond machinery makes visible.*

Table 3: Foveation vs. uniform at $\sigma_{\max} = 8$ Gaussian blur with the same ResNet-18 encoder. Same on H1, different on H2 / behavioural / generalisation. Sources: Table 1, Figure 3, Figure 5, Figure 4. Data are deterministic-rollout single-seed.

Measure	Foveated (fix)	Uniform	Verdict
LSTM GPS R^2 (Gibson)	$+0.06 \pm 0.88$	-0.31 ± 0.86	both chance, *indistinguishable*
Encoder $\to$ GPS R^2	-0.65 ± 0.28	-0.32 ± 0.08	both negative, overlapping
LSTM compass R^2 (Gibson)	$+0.07 \pm 0.69$	$+0.36 \pm 0.23$	uniform $\sim 5\times$ better
LSTM GPS R^2 (MP3D, held out)	$+0.35 \pm 0.27$	-0.36 ± 1.38	**differ by 0.71**
Shortcut SPL drop %	21%	41%	**uniform $2\times$ more brittle**
Memory-transplant cost (fov $\to$ uni)	-0.10 SPL beyond self-noise		**not interchangeable**
Pairwise CKA(fov, uni)	$< 10^{-4}$		**near-orthogonal subspaces**
1-NN purity (pooled)	1.000 vs. chance 0.20		**linearly separable**

Four in-flight experiments target specific predictions of the encoder–memory race within the foveation slot (designs in Appendix G). One result has landed and is pivotal:

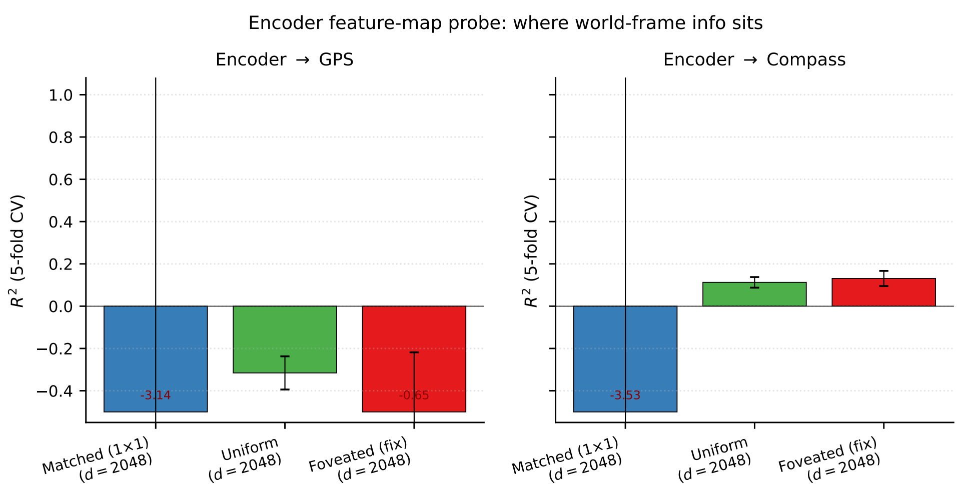


Figure 4: Encoder feature-map probe (post ResNet-18, pre-LSTM, 5-fold CV Ridge). Encoder dim $d = 2048$ (compressed flatten) for all three sighted conditions. None of the three encoders linearly decode GPS (matched $R^2 = -3.14$, uniform -0.32, foveated -0.65); the encoder–memory race is therefore not about encoder-level GPS information availability. Compass shows a small positive on uniform / foveated ($+0.11, +0.13$); matched compass is also chance.

Encoder feature-map probe: where world-frame info actually sits Probing the visual encoder's output directly (post ResNet-18, pre-LSTM, via forward-hook on the compression layer) gives a striking result: *none of the three sighted encoders linearly decode GPS* (Figure 4). Compass also fails or is well below ceiling. *This sharpens the H1 mechanism*: the divergence happens inside the LSTM, not at the visual encoder. Bottleneck conditions integrate the GPS-sensor input across time because their visual stream gives the LSTM minimal feature variety to substitute; rich-encoder conditions have richer (though still position-blind) visual features that the LSTM uses instead. The "LSTM gain" — the difference between top-layer LSTM GPS R^2 and the encoder's — is a clean signature: matched gains $+3.9$ R² (LSTM heavily reconstructs from sensor integration), uniform gains $+0.0$ (LSTM fully substitutes visual features for GPS), and foveated gains $+0.7$ (LSTM partially reconstructs — consistent with foveation supplying *less* navigation-useful visual structure than uniform, so the LSTM cannot rely on visual features alone).

Provisional conclusion. Foveation under $\sigma_{\max} = 8$ Gaussian blur is *not a copy of uniform*: it converges on H1 but diverges on H2 / behaviour / dataset-shift (Table 3). The encoder feature-map probe further shows the encoder discards linear GPS regardless of condition, so *whatever foveation does differently from uniform, it does it inside the LSTM in subspaces invisible to a linear-GPS probe*. The four in-flight experiments (Appendix G) test *what* that "something different" is. F3 log-polar in particular has a falsifiable prediction: log-polar's $\sim 2{\times}2$ encoder spatial output should produce LSTM GPS $R^2 \geq 0.3$ (between matched's $+0.78$ and uniform's ≈ 0); if instead it matches uniform, the encoder-spatial-output mechanism would need reframing.

4.5 Probe-readable vs. policy-used memory: a $2{\times}2$ dissociation

H1 (§4.2) measures what is *readable* from the recurrent state. The shortcut-discovery experiment measures what the policy actually *relies on*: in paired same-scene/different-goal episodes (20 scenes $\times$ 10 pairs), the second-episode SPL drop under persistent vs. reset memory indexes how tightly the policy couples actions to integrated recurrent memory (blind loses 52%, uniform 41%, foveated 21%, matched 18%, foveated-learned 15%). Plotting probe-readable GPS against this behavioural reliance reveals a $2{\times}2$ pattern (Figure 5) with two opposite anomalies. Matched-compute has a linear top-layer GPS code yet its policy weights it less than the probe predicts; uniform has no linear GPS but relies heavily on memory — plausibly via non-spatial scene-history features. "Probe-readable" and "policy-relied-on" can therefore dissociate in either direction.

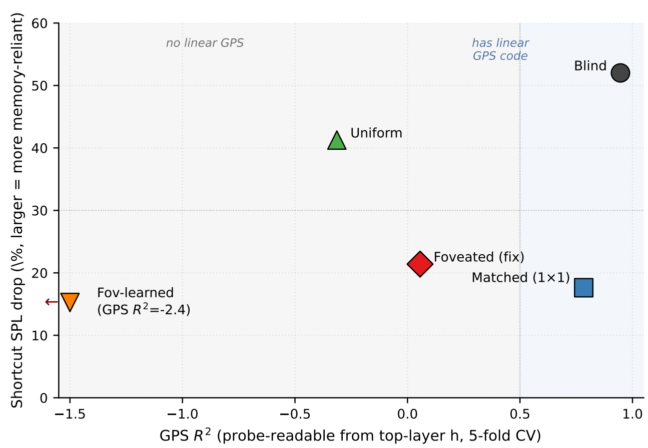


Figure 5: Probe-readable GPS (H1) vs. behavioural memory reliance (shortcut). The five conditions occupy four corners of a $2{\times}2$: blind "has GPS, uses memory" (consistent with H1); foveated-fixed and foveated-learned "no linear GPS, low memory reliance" (also consistent with H1); **matched** and **uniform** are the two off-diagonal anomalies. Fov-learned's $R^2 = -2.43$ is clipped at -1.5.

A trajectory-level look at *how* persistent-memory failures manifest sharpens the dissociation (Figure 6). Quantifying across all same-floor paired-episode failures, only uniform's persistent agent ends up closer to the previous-episode goal than to the new one (Table 4, margin $+1.83$m, n=46); blind's persistent runs end nearer the new goal but never reach it (margin -0.38m). This pattern is consistent with the $2{\times}2$: *uniform's memory acts as a visual / scene-history anchor* that locks the agent in place, whereas *blind's GPS code interferes via a different channel* (the recurrent state mis-reporting current position to the policy's goal-direction integrator). The single-seed nature of these claims is a limitation; multi-seed replication is in flight.

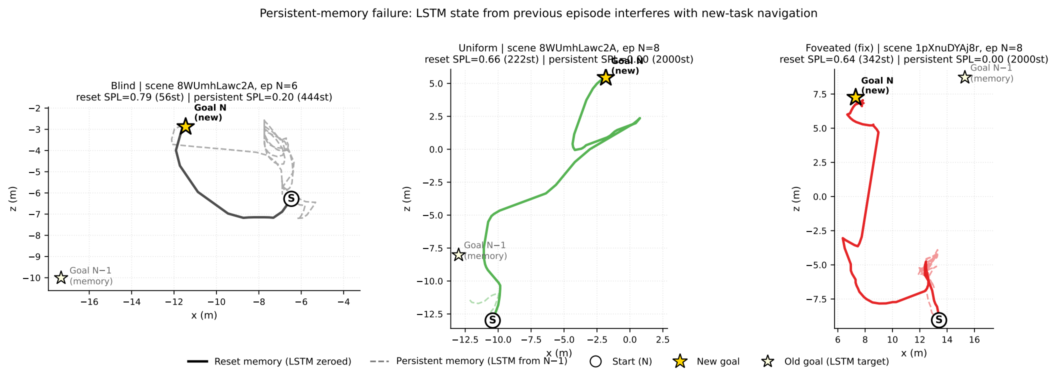


Figure 6: Three persistent-memory failure modes (one paired-episode case per condition). *Blind*: persistent agent attempts the new goal, loops, gives up at 443 steps. *Uniform*: persistent agent stays *at* the old-goal location for the full 2000-step cap. *Foveated*: persistent agent loops chaotically in a region distant from both goals.

Table 4: Where the persistent agent ends up (same-floor paired-episode failures: reset SPL > 0.5, persistent SPL < 0.2, persistent steps $\geq$ 30, $|\Delta y_{\text{goal}}| < 0.5$m). Only uniform's failures cluster nearer the previous-episode goal.

Condition	n	avg dist $\to$ old (m)	avg dist $\to$ new (m)	margin (m)	Closer to	Reading
Blind	27	7.48	7.10	-0.38	new	tries new, fails
Matched	35	6.29	5.72	-0.57	new	tries new, fails
Uniform	46	**6.45**	8.28	$\mathbf{+1.83}$	**old**	locks onto old
Foveated (fix)	16	7.01	6.42	-0.59	new	wanders
Fov-learned	5	3.89	6.19	$+2.30$	old	(n=5, low confidence)

4.6 Gaze location (H3, in progress)

H3 asks whether gaze *location* acts as a second representational-content axis within the rich-encoder regime. Our minimal learned-gaze module (deterministic sigmoid MLP, slow updates per 128-step segment) freezes at $(0.49, 0.62)$ with per-dimension std $\leq 2.5 \times 10^{-4}$ after 250M frames; anti-collapse regularisers restore variance only at task-cost (Appendix E). The collapse is not evidence about active gaze in general — the decoder is minimal by design — but it provides an empirical anchor for the gaze-location test: a foveated-shifted control with gaze hardcoded at $(0.49, 0.62)$ instead of $(0.5, 0.5)$, identical to foveated-fixed in every other respect, isolates static gaze location from the optimisation-path interaction that produced the collapsed gaze. Training is in progress at submission.

4.7 Boundaries

Three probes that do *not* discriminate define the boundary of our claims. Coarse occupancy decodes above chance everywhere; metric 20×20 occupancy fails everywhere (consistent with [Ramakrishnan et al., 2025] that high-resolution metric layout needs non-linear probes); the ego-frame goal vector is at chance everywhere because PointGoal supplies it as a per-step input. Population coding (Appendix D) sharpens H1 in a direction we did not expect: rich-encoder conditions have a few units with *higher* peak spatial information than blind, but those peaked units do not decode GPS — suggesting they encode features merely correlated with position (visual landmarks, trajectory phase). Blind's broader, lower-peak distribution carries the actual linear position code.

5 Discussion

5.1 Two orthogonal axes: sensor structure and gaze location

Our findings unify under a candidate account we call the *encoder–memory race*: top-layer linear GPS encoding in the LSTM is *maintained* when the encoder hands the LSTM minimal spatial-feature variety per step (blind, matched-compute), and *overwritten* when richer visual features substitute (uniform, foveated, foveated-learned). The encoder feature-map probe (§4.4) confirms that no sighted encoder linearly re-derives GPS, so the H1 race is not encoder-vs-LSTM at the position-information level; it is at whether the LSTM has visual variety enough to substitute for integrating its Layer-0 GPS-sensor signal across time. H2 (§4.3) cuts orthogonally: *what* the LSTM encodes lives in distinct linear subspaces across conditions, and the difference matters for behaviour (cross-condition memory transplants drop SPL beyond transplant-mechanics noise; the 5×5 matrix is asymmetric). H3 (§4.6) asks whether, within the rich-encoder pass-through regime, gaze location is a second content axis; the foveated-shifted causal control is in training. We present the unifying account as a candidate, not a proven mechanism: a direct causal test (e.g. ablating the visual stream mid-rollout) is left for follow-up.

5.2 Biological precedent: sensor structure shapes hippocampal representation

The principle that sensor structure constrains spatial-memory format is not one we propose; biological evidence for it predates ours by decades. Our contribution is to isolate the principle in a controlled silicon model where the encoder, memory, policy, training distribution and reward are held fixed and only the visual sensor varies. Three threads of animal-navigation research are particularly close to the three findings.

H1 (sensor bottleneck $\rightarrow$ memory takes over). Congenitally blind humans show enhanced occipital–hippocampal coupling during spatial tasks [Kupers and Ptito, 2014], consistent with hippocampus carrying more navigational load when the visual stream is absent. Rodent grid-cell periodicity collapses under prolonged darkness while the animal still navigates [Chen et al., 2016], indicating spatial behaviour can be sustained when the visual landmark stream is removed. Both are biological analogs of our blind / matched-compute regime, where the LSTM — deprived of rich visual variety per step — retains a linear top-layer GPS code that the rich-encoder conditions overwrite.

H2 (different sensors $\rightarrow$ different spatial codes). The primate vs. rodent contrast in hippocampal cell types — view-cells dominant in primates, place-cells dominant in rodents — has been attributed in part to the foveated-with-saccades vs. near-panoramic visual sampling regime each species uses [Rolls, 2024]. Cross-modality remapping in echolocating bats, where the same neurons recode under vision vs. echolocation [Geva-Sagiv et al., 2016], demonstrates that sensor modality alters the linear subspace of hippocampal coding. Both parallel our H2 finding: the same task with different visual structure yields linearly separable LSTM subspaces (CKA $< 10^{-4}$, probe-transfer collapse, asymmetric memory transplant).

H3 (gaze location modulates memory encoding). In primates, hippocampal theta is gated by saccades, and the reliability of this saccade-gated signal predicts memory encoding success [Jutras et al., 2013, Tas et al., 2016]. Where the eye points is therefore not memory-neutral. Whether the analogous prediction holds for an LSTM gaze module is what §4.6 is testing (foveated-shifted causal control in training).

We treat these as convergent (not conclusive) evidence. Convergence across systems — and across silicon vs. neural — is suggestive of a sensor–memory coupling principle more general than any single architecture, but it does not show that the mechanisms are the same. Our experiments isolate the principle in a controlled, single-architecture setting; biological work establishes that the principle exists in animals at all. The two are complementary; neither is decisive alone.

5.3 Why this, not alternative accounts

Four alternative explanations are ruled out: skill difference (success bunched 93–99%), probe capacity (DtG decodes everywhere), sample-size or stochastic-sampling artefacts (Appendix A), and linear-probe artefact (memory transplant reproduces the ordering outside the probe framework; MLP probe shows the divergence is specific to *linear* top-layer decodability, which is what the linear DD-PPO action head reads). Full rebuttals in Appendix C.

5.4 Implications and scope

If the encoder–memory race is causal — the resolution scaling sweep (Appendix H) tests this — encoder capacity becomes a practical training-time lever for inducing cognitive-map-like representations. The relevant axis is not pixel count: matched-compute's 48^2 input with a 1×1 feature map has a strong linear top-layer GPS code while uniform's 256^2 does not; we attribute the pattern to encoder spatial-feature *variety* per step. Outside the recurrent setting we test, H3 suggests a prediction for transformer navigators: if learned attention selects observation content the way our gaze module does, strong constraints on attended tokens should produce content-level dissociations analogous to ours. Whether sensor–memory coupling extends to supervised visual learners, non-visual modalities, or multimodal systems is open.

5.5 Limitations

(i) *Single seed* for the H1 numbers; multi-seed replication is in progress and gates the headline claims. (ii) *Foveated-fixed checkpoint* is ckpt.36 ($\sim$174M frames, last clean before a silent NaN-gradient corruption; Appendix F); the other four conditions use the final converged checkpoint. (iii) *Learned-gaze module* is a minimal deterministic sigmoid MLP that collapsed under pure task supervision; the H3 test (foveated-shifted control) does not depend on this module working. Richer gaze mechanisms (stochastic gaze, intrinsic-reward gaze, attention-based selection) are out of scope. (iv) *Foveation model:* $\sigma_{\max} = 8$ Gaussian blur is a simplification; a log-polar control and an $\sigma_{\max} \in \{2, 4, 12, 20\}$ strength sweep are in training (Appendix G). (v) *Linear probes only:* non-linear probes might reveal structure Ridge misses (a 2-layer MLP partially recovers GPS in rich-encoder conditions; Appendix B).

6 Conclusion

A controlled five-condition ablation, with the visual pathway as the only varied component, identifies sensor structure as a representational-content axis in recurrent navigation agents: the encoder–memory race (H1), condition-specific representational format (H2), and a probe-readable–vs.–policy-used dissociation that emerges at the H1$\times$shortcut intersection. The pattern reproduces on held-out MP3D and parallels independent animal-navigation findings, suggesting a sensor–memory coupling principle that our experiments do not show is architecture-independent. Open: a finer-grained encoder-resolution sweep (Appendix H) for causal H1; the foveated-shifted control for H3; whether transformer navigators reproduce the race; and multi-seed replication of the H1 numbers.

References

Andrei Atanov, Yicheng Fu, Gaurav Singh, Jianren Yu, Andrew Spielberg, and Amir Zamir. How far can a 1-pixel camera go? solving vision tasks using photoreceptors and computationally designed visual morphology. In *European Conference on Computer Vision (ECCV)*, 2024.

Andrea Banino, Caswell Barry, Benigno Uria, Charles Blundell, Timothy Lillicrap, Piotr Mirowski, Alexander Pritzel, Martin J Chadwick, Thomas Degris, Joseph Modayil, et al. Vector-based navigation using grid-like representations in artificial agents. *Nature*, 557(7705):429–433, 2018.

Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.

Guifen Chen, John A. King, Neil Burgess, and John O'Keefe. Absence of visual input results in the disruption of grid cell firing in the mouse. *Current Biology*, 26(17):2335–2342, 2016.

Christopher J Cueva and Xue-Xin Wei. Emergence of grid-like representations by training recurrent neural networks to perform spatial localization. In *International Conference on Learning Representations (ICLR)*, 2018.

Kshitij Dwivedi, Gemma Roig, Aniruddha Kembhavi, and Roozbeh Mottaghi. What do navigation agents learn about their environment? In *Computer Vision and Pattern Recognition (CVPR)*, 2022.

Maya Geva-Sagiv, Sandro Romani, Liora Las, and Nachum Ulanovsky. Hippocampal global remapping for different sensory modalities in flying bats. *Nature Neuroscience*, 19(7):952–958, 2016.

James Gornet and Matt Thomson. Automated construction of cognitive maps with visual predictive coding. *Nature Machine Intelligence*, 6:820–833, 2024.

Jay A Hennig, Sandra A Romero Pinto, Takahiro Yamaguchi, Scott W Linderman, Naoshige Uchida, and Samuel J Gershman. Emergence of belief-like representations through reinforcement learning. *PLOS Computational Biology*, 19(9):e1011067, 2023.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

Michael J. Jutras, Pascal Fries, and Elizabeth A. Buffalo. Oscillatory activity in the monkey hippocampus during visual exploration and memory formation. *Proceedings of the National Academy of Sciences*, 110(32):13144–13149, 2013.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning (ICML)*, 2019.

Ron Kupers and Maurice Ptito. Compensatory plasticity and cross-modal reorganization following early visual deprivation. *Neuroscience & Biobehavioral Reviews*, 41:36–52, 2014.

Volodymyr Mnih, Nicolas Heess, Alex Graves, and Koray Kavukcuoglu. Recurrent models of visual attention. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.

Motahareh Pourrahimi and Pouya Bashivan. Emergent brain-like representations in a goal-directed neural network model of visual search. *bioRxiv preprint*, 2025.

Santhosh Kumar Ramakrishnan, Erik Wijmans, Philipp Krähenbühl, and Vladlen Koltun. Does spatial cognition emerge in frontier models? In *International Conference on Learning Representations (ICLR)*, 2025.

Edmund T. Rolls. Why do primates have view cells instead of place cells? *Trends in Cognitive Sciences*, 28(9):862–874, 2024.

Ruth Rosenholtz. Capabilities and limitations of peripheral vision. *Annual Review of Vision Science*, 2:437–457, 2016.

Manolis Savva, Abhishek Kadian, Oleksandr Maksymets, Yili Zhao, Erik Wijmans, Bhavana Jain, Julian Straub, Jia Liu, Vladlen Koltun, Jitendra Malik, Dhruv Batra, and Devi Parikh. Habitat: A platform for embodied AI research. In *International Conference on Computer Vision (ICCV)*, 2019.

Jinghuan Shang and Michael S. Ryoo. Active vision reinforcement learning under limited visual observability. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Nathan Sprague and Dana H. Ballard. Eye movements for reward maximization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2003.

Hans Strasburger, Ingo Rentschler, and Martin Jüttner. Peripheral vision and pattern recognition: A review. *Journal of Vision*, 11(5):13, 2011.

Amanda C. Tas, Steven J. Luck, and Andrew Hollingworth. The role of attention in the encoding of visual working memory. *Journal of Experimental Psychology: Human Perception and Performance*, 42(8):1121–1138, 2016.

Erik Wijmans, Abhishek Kadian, Ari Morcos, Stefan Lee, Irfan Essa, Devi Parikh, Dhruv Batra, and Manolis Savva. DD-PPO: Learning near-perfect pointgoal navigators from 2.5 billion frames. In *International Conference on Learning Representations (ICLR)*, 2020.

Erik Wijmans, Manolis Savva, Irfan Essa, Stefan Lee, Ari Morcos, and Dhruv Batra. Emergence of maps in the memories of blind navigation agents. In *International Conference on Learning Representations (ICLR)*, 2023.

Sylvia Wirth, Pierre Baraduc, Angélique Planté, Serge Pinede, and Jean-René Duhamel. Gaze-informed, task-situated representation of space in primate hippocampus during virtual navigation. *PLOS Biology*, 15(2):e2001045, 2017.

Fei Xia, Amir R Zamir, Zhiyang He, Alexander Sax, Jitendra Malik, and Silvio Savarese. Gibson Env: Real-world perception for embodied agents. In *Computer Vision and Pattern Recognition (CVPR)*, 2018.

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A Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4-step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

Length-matching ablation. Under deterministic rollouts, episode lengths span ∼100–400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. **[TODO: A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect it to leave the H1 ordering unchanged because the bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.]**

Cross-heading probe. The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes; not in our results.

B Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch. The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder's flattened output, and feeds the result as the LSTM's Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 2c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details. For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states + GPS labels as the linear Ridge probe, with the same 5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, foveated-learned 0.73, uniform 0.55 (vs. Ridge's chance R²). The MLP probe alone cannot disentangle "non-linear position code" from "MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)"; the resolution scaling sweep (Appendix H) tests this more directly by varying input resolution at fixed encoder stack.

C Alternative-account rebuttals (full)

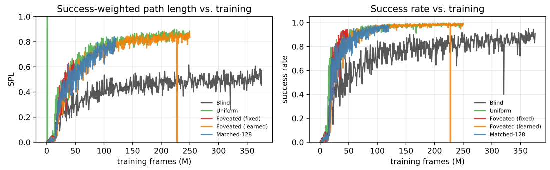
Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all five conditions ($R^2 \geq 0.81$), demonstrating Ridge has the capacity to fit high-R^2 scalars when the hidden state contains them. Hewitt–Liang label-permutation selectivity is high where GPS is high (blind, matched) and at chance where GPS is at chance — the latter follows from real and control probes facing the same near-zero target-information ceiling. The chance-level rich-encoder R^2 is therefore a hidden-state property, not a probe artefact.

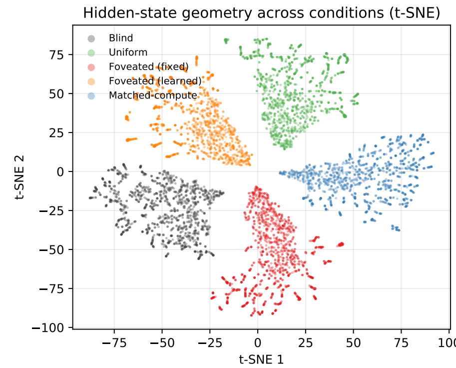
Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with $\sim$ 100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix A.

Linear-probe artefacts. Two complementary checks. *(a) Behavioural:* the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-pass-through ordering; the 5×5 matrix's asymmetry adds a structure no linear-similarity measure can read. *(b) Non-linear probe:* a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix B). The bottleneck-vs-pass-through divergence is therefore not about whether *any* probe can recover position; it is specifically about whether a *linear* top-layer probe can. Since the DD-PPO action head is a linear projection from $\mathbf{h}_2$, the linear metric is what the policy itself can use, so H1 maps onto policy-accessible encoding regardless of how the MLP interpretation resolves.

D Supplementary visualisations



(a) Training curves (TensorBoard event-file scalars). All four sighted conditions converge to SPL 0.82–0.85 and success $> 97\%$ by $\sim$100M frames; blind converges to SPL $\sim$0.50 over a longer horizon. The glitch at frame $\approx$ 228M (fov-learned) is a resume-boundary artefact in the scalar log.



(b) t-SNE projection of top-layer LSTM hidden states (deterministic-rollout probes), pooled across all five conditions (perplexity 30). The five conditions occupy non-overlapping regions of the 2-D embedding — a visual confirmation of the near-zero CKA result (§4.3).

Figure 7: Supplementary visualisations referenced from the main text.

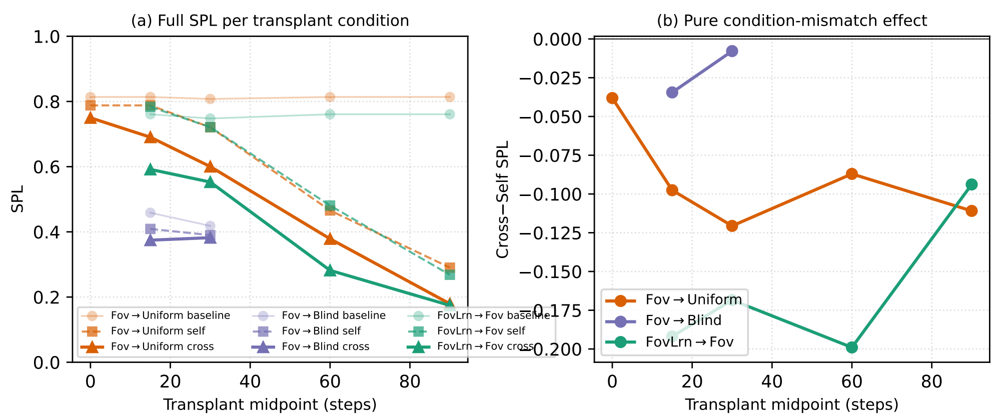


Figure 8: Memory-transplant midpoint sweep on three representative donor→recipient pairs (foveated→uniform, foveated→blind, foveated-learned→foveated) across midpoints $\{0, 15, 30, 60, 90\}$ time-steps (n=100–150 episodes/cell). The full 5×5 matrix in Figure 3 (right) is at midpoint t=30; this supplementary figure shows that the cross-self gap is roughly stable for $t \geq 15$ and that the t=0 self-/cross- baseline collapses to a protocol-noise floor (so the diagonal in Figure 3 is correctly read as "rollout-divergence noise" rather than "condition-mismatch noise"). Foveated-learned→foveated incurs -0.17 SPL on average across $t \geq 15$, foveated→uniform -0.10, foveated→blind -0.02, consistent with the asymmetry pattern in the full matrix.

E Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation. In the first foveated-learned training run the gaze decoder collapsed (per-dimension std $\approx 5 \times 10^{-4}$). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative $\sigma'(x) \to 0$ attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

Target-variance hinge. We add $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma^2_{\text{target}} - \text{Var}_{\text{batch}}[g])$ where $g \in [0,1]^2$, $\text{Var}_{\text{batch}}$ averaged over the two gaze dimensions, $\sigma^2_{\text{target}} = 0.01$ (std ≈ 0.1), $\lambda = 0.01$. The hinge disappears once variance exceeds the target. We also evaluated an un-capped $-\lambda \cdot \text{Var}[g]$ variant as a negative control.

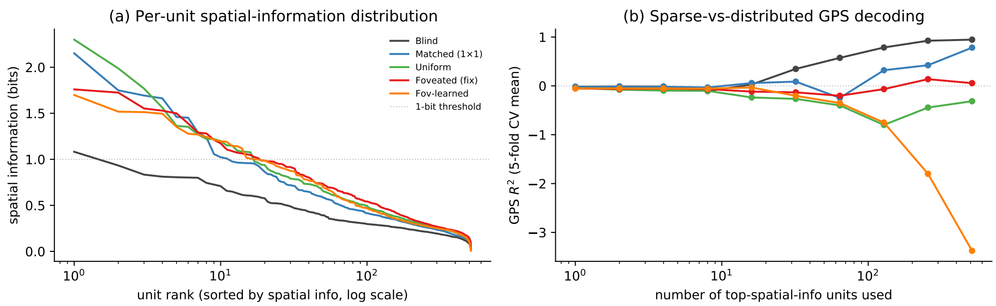


Figure 9: Population coding analysis (referenced from §4.7). *(a)* Per-unit spatial-information distribution: each curve is one condition's units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform / foveated / fov-learned) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). *(b)* Sparse-vs-distributed GPS decoding: probe R^2 when only the top-k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; matched climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information* $\neq$ *position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

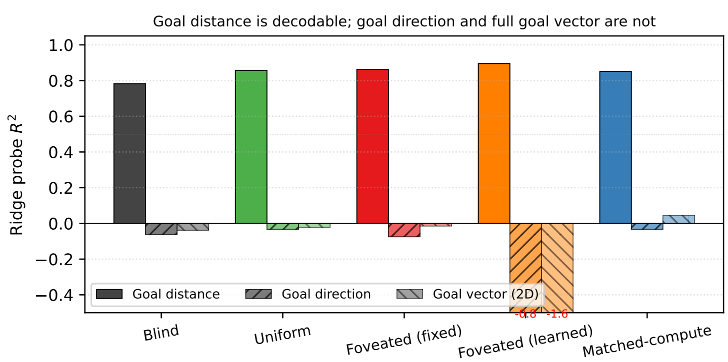
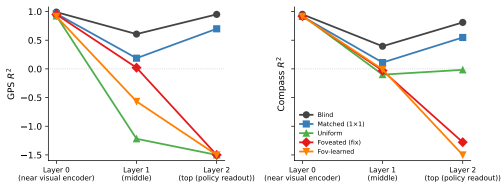


Figure 10: Goal-vector probe R^2 (referenced from §4.7). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$, validating probe machinery); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

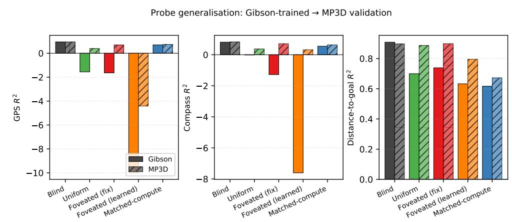
Pilot 1 (uncapped, job 2840256). After 10 M frames, gaze variance exceeded that of a uniform distribution over $[0, 1]^2$ (std $[0.43, 0.42]$ vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at ~ 8 M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

Pilot 2 (hinge, job 2842288). Gaze converged to an extreme corner of image space: mean $(0.001, 0.998)$ with std $[0.020, 0.023]$ — well below the target std of 0.1 and therefore with the hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

Interpretation. Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze location with entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an



(a) Per-LSTM-layer GPS / compass probe R^2 across all three layers, all five conditions (single train/test fit; the 5-fold CV mean reported in Table 1 matches qualitatively but with different magnitudes for rich-encoder conditions). Compass tells the same story as GPS at smaller magnitude.



(b) Companion to Figure 2d (which uses 5-fold CV mean): single train/test fit on the same checkpoints; the qualitative bottleneck-vs-pass-through partition is the same.

Figure 11: Per-layer and MP3D companion figures (referenced from §4.2).

architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

F Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈170M frames of stable progress (reward ≈ 10, SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL$= 0$ until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PPO.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.

G Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from §4.4. All four are submitted; numbers will populate §4.4 in revision as they land.

Table 5: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	in training
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training
Encoder feature-map probe (matched / uniform / foveated)	where position information is held	**landed; see §4.4**
Foveated-shifted (gaze $(0.49, 0.62)$)	gaze-location axis (links to H3)	in training

Foveation strength sweep (F1–F4). We train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$

it should approach matched-compute. The R^2 vs. $\sigma_{\max}$ curve directly tests the encoder–memory race as a continuous lever rather than a binary one.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 8×8 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2\times2$, much closer to matched's 1×1 than to uniform's 8×8. This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder output*. Falsifiable prediction (written before result available): log-polar should produce LSTM GPS $R^2 \geq 0.3$, between matched's $+0.78$ and uniform's ≈ 0, with corresponding behavioural shifts. If instead log-polar matches uniform, the encoder-spatial-output-dimensionality mechanism is wrong. F3 in training (~14M / 250M frames at submission, ETA 2–3 days).

Foveated-shifted (gaze location). Training a foveated agent with hardcoded static gaze at $(0.49, 0.62)$ matches foveated-learned's collapsed gaze location while removing the learned-gaze module from the optimisation path. If foveated-shifted does *not* reproduce foveated-learned's specific behaviours (Gibson–MP3D compass swing, transplant ranking), those behaviours can be attributed to the optimisation-path interaction rather than to gaze location alone. Links to H3 (§4.6).

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists vs. what is in flight, and (ii) the reader can identify which incoming result would change the paper's interpretation.

H Encoder-capacity scaling sweep

[TODO: The scaling sweep trains matched-compute variants at input resolutions $\{32, 48, 64, 96, 128, 192\}$ alongside the existing blind (no encoder) and uniform 256×256 anchors. Probe results on the trained checkpoints will populate Figure 12 and this appendix section. Hypothesis: GPS probe R^2 decreases monotonically with input resolution, from $R^2 \approx 0.95$ at no-encoder to $R^2 \approx 0$ at 256×256, crossing into the pass-through regime between 48 and 128 pixels.]

Figure 12: Encoder-capacity scaling curve. **[TODO: Pending data.]** x-axis: input resolution in pixels; y-axis: GPS and compass probe R^2 (5-fold CV, mean$\pm\sigma$).